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# Conformal prediction under ambiguous ground truth

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## Abstract

Conformal Prediction (CP) allows to perform rigorous uncertainty quantification by constructing a prediction set $C(X)$ satisfying $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$ for a user-chosen $\alpha \in [0, 1]$ by relying on calibration data $(X_1, Y_1), ..., (X_n, Y_n)$ from $\mathbb{P} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}$. It is typically implicitly assumed that $\mathbb{P}^{Y|X}$ is the "true" posterior label distribution. However, in many real-world scenarios, the labels $Y_1, ..., Y_n$ are obtained by aggregating expert opinions using a voting procedure, resulting in a one-hot distribution $\mathbb{P}^{Y|X}_{\text{vote}}$. This is the case for most datasets, even well-known ones like ImageNet. For such "voted" labels, CP guarantees are thus w.r.t. $\mathbb{P}_{\text{vote}} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}_{\text{vote}}$ rather than the true distribution $\mathbb{P}$. In cases with unambiguous ground truth labels, the distinction between $\mathbb{P}_{\text{vote}}$ and $\mathbb{P}$ is irrelevant. However, when experts do not agree because of ambiguous labels, approximating $\mathbb{P}^{Y|X}$ with a one-hot distribution $\mathbb{P}^{Y|X}_{\text{vote}}$ ignores this uncertainty. In this paper, we propose to leverage expert opinions to approximate $\mathbb{P}^{Y|X}$ using a non-degenerate distribution $\mathbb{P}^{Y|X}_{\text{agg}}$. We then develop *Monte Carlo CP* procedures which provide guarantees w.r.t. $\mathbb{P}_{\text{agg}} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}_{\text{agg}}$ by sampling multiple synthetic pseudo-labels from $\mathbb{P}^{Y|X}_{\text{agg}}$ for each calibration example $X_1, ..., X_n$. In a case study of skin condition classification with significant disagreement among expert annotators, we show that applying CP w.r.t. $\mathbb{P}_{\text{vote}}$ under-covers expert annotations: calibrated for 72% coverage, it falls short by on average 10%; our Monte Carlo CP closes this gap both empirically and theoretically. We also extend Monte Carlo CP to multi-label classification and CP with calibration examples enriched through data augmentation.

## 1 Introduction

Many application domains, especially safety-critical applications such as medical diagnostics, require reasonable uncertainty estimates for decision making and benefit from statistical performance guarantees. Conformal prediction (CP) is a statistical framework allowing to quantify uncertainty rigorously by providing finite-sample, non-asymptotic performance guarantees. First introduced by Vovk et al. (2005), it has become very popular in machine learning as it is widely applicable while making no dsitributional or model assumptions, e.g., see (Romano et al., 2019; Sadinle et al., 2019; Romano et al., 2020; Angelopoulos et al., 2021; Stutz et al., 2021; Fisch et al., 2022) and Angelopoulos & Bates (2021) for more references.

Specifically, we consider a classification task with $K$ classes and denote by $[K]$ the set $\{1, \dots, K\}$. Then, a classifier $\pi : \mathcal{X} \to \Delta^K$ outputs the class probabilities where $\Delta^K$ is the $K$-simplex. Based only on a held-out set of $n$ calibration examples $(X_i, Y_i) \sim \mathbb{P}$, CP allows us to return a prediction set $C(X) \subseteq [K]$ for a given test point $(X, Y)$ (with $Y$ being unobserved) dependent on the calibration data such that

$$\mathbb{P}(Y \in C(X)) \geq 1 - \alpha, \tag{1}$$

whatever being $\mathbb{P}$ and $\pi$ as long as the joint distribution of $((X_1, Y_1), \dots, (X_n, Y_n), (X, Y))$ is exchangeable. Here $\alpha \in [0, 1]$ is a user-specified parameter and the probability in Equation (1) is not only over $(X, Y)$ but also over the calibration set. This is called the *coverage guarantee* of CP. The size of such prediction sets $|C(X)|$, also called *inefficiency*, is a good indicator of the uncertainty for $X$.

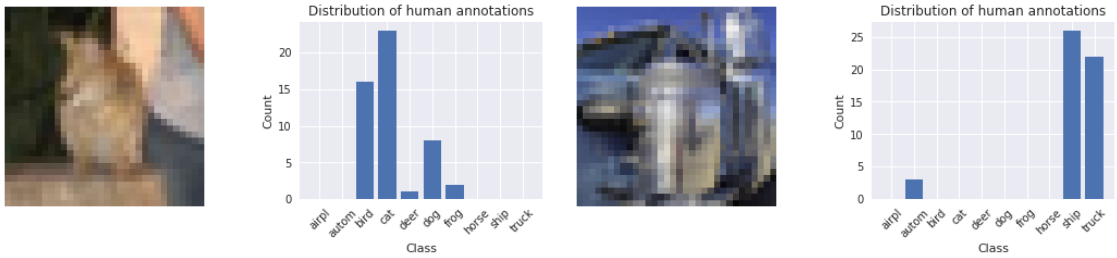


Figure 1: Two ambiguous examples from CIFAR10-H (Peterson et al., 2019) with the corresponding annotations of $p = 50$ experts. These annotations can be used to build $\mathbb{P}_{\text{agg}}(Y = y|X = x, Y^1, ..., Y^p)$, see text for details.

It is commonly taken for granted that $\mathbb{P} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}$ where $\mathbb{P}^{Y|X}$ is the "true" posterior label distribution so that the l.h.s. of Equation (1) is the probability for the ground truth label $Y$ being an element of the prediction set $C(X)$. However, in many practical applications, the calibration labels $(Y_i)_{i\in[n]}$ are obtained based on (multiple) expert opinions. In medical applications, for example, it is usually impossible to identify the patient's actual condition as this would require invasive and expensive tests. Instead, labels are derived from expert annotations, e.g., doctor ratings (Liu et al., 2020). Beyond medical diagnosis, however, this is generally the case for many popular datasets such as CIFAR10 (Krizhevsky, 2009; Peterson et al., 2019), COCO (Lin et al., 2014), ImageNet (Russakovsky et al., 2015) or many datasets from the GLUE benchmark (Wang et al., 2019) such as MultiNLI (Williams et al., 2018) (see Appendix A). Based on the expert annotations, the majority voted label is then typically selected as the ground truth label. Formally, this means that we consider $Y_i \sim \mathbb{P}_{\text{vote}}(\cdot|X = x_i)$ where $\mathbb{P}_{\text{vote}}(Y = y|X = x)$ is a one-hot distribution. Thus, when using CP on these *voted labels*, we obtain guarantees of the form $\mathbb{P}_{\text{vote}}(Y \in C(X)) \geq 1 - \alpha$ for $\mathbb{P}_{\text{vote}} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}_{\text{vote}}$ rather than a guarantee w.r.t. to the true distribution, $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$. The difference between $\mathbb{P}_{\text{vote}}$ and $\mathbb{P}$ is small for some tasks, including popular benchmarks such as CIFAR10, because most examples are fairly unambiguous, meaning that annotators do rarely disagree because $\mathbb{P}(Y = y|X = x)$ is one-hot anyway and a simple aggregation strategy can unambiguously determine the true class. However, in complex tasks such as the dermatology problem addressed in this work, there are many examples where experts disagree. More importantly, this disagreement is often "irresolvable" (Schaekermann et al., 2016; Gordon et al., 2022; Uma et al., 2022), meaning that obtaining more annotations will likely not reduce disagreement. In such cases, $\mathbb{P}_{\text{vote}}(Y = y|X = x)$ differs significantly from the true conditional distribution $\mathbb{P}(Y = y|X = x)$.

In such scenarios, performing CP using voted labels can severely underestimate the true uncertainty (see Section 2.2 for an intuitive example). Instead of summarizing the expert annotations by a single one-hot distribution, we propose here to rely on an approximation of $\mathbb{P}_{\text{agg}}(Y = y|X = x)$ to $\mathbb{P}(Y = y|X = x)$ to better account for this uncertainty. As a simple example, assume that for each calibration data $X$, expert $q \in [p]$ annotates a single class $Y^q \in [K]$ as done on CIFAR10-H (Peterson et al., 2019), see Figure 1 for an illustration. Then, let $p_k = \sum_{q=1}^{p} \mathbb{I}(Y^q = k)$ be the number of experts selecting class $k$, a simple voting procedure sets $\mathbb{P}_{\text{vote}}(Y = y|X = x, Y^1, ..., Y^p) = \mathbb{I}(y = \hat{Y})$ where $\hat{Y} = \arg\max_{k\in[K]} \; p_k$ (disregarding ties for simplicity) and unconditionally we also typically will have $\mathbb{P}_{\text{vote}}(Y = y|X = x)$ a one-hot distribution in unambiguous cases (i.e. the same label is selected for all expert annotations). Obviously, if there is sufficient disagreement among annotators, taking $\hat{Y}$ will ignore many of the (usually expensive) annotations. As a result, performing CP on the voted labels ignores this uncertainty. Instead, we argue that selecting an aggregated distribution, e.g., $\mathbb{P}_{\text{agg}}(Y = y|X = x, Y^1, ..., Y^p) = \frac{1}{p}\sum_{k\in[K]} p_k \mathbb{I}(y = k)$, allows us to better capture uncertainty present in the annotations. Ideally, by integrating over $(Y^1, ..., Y^p)$, the resulting $\mathbb{P}_{\text{agg}}$ is a good approximation of the true distribution $\mathbb{P}$ (and we discuss common ways of constructing $\mathbb{P}_{\text{agg}}$ in Section 3.1).

**Contributions:** In this paper, we propose CP procedures that allow us to construct a prediction set $C(X)$ satisfying $\mathbb{P}_{\text{agg}}(Y \in C(X)) \geq 1-\alpha$ for $\mathbb{P}_{\text{agg}} = \mathbb{P}^X \otimes \mathbb{P}_{\text{agg}}^{Y|X}$ as long as one can sample from $\mathbb{P}_{\text{agg}}^{Y|X}$[1]. Note that while it would be desirable to have instead guarantees w.r.t. the distribution $\mathbb{P}$, this is an impossible task as we never observe any data with labels sampled from $\mathbb{P}^{Y|X}$. Whether $\mathbb{P}_{\text{agg}}^{Y|X}$ is a good approximation to $\mathbb{P}^{Y|X}$ will be application dependent. Instead, the prediction set $C(X)$ outputted by our procedures is the one we could compute if the experts had access to $X$ and their opinions were aggregated through $\mathbb{P}_{\text{agg}}^{Y|X}$. This is the best one can hope for. We emphasize that we are not making more model assumptions than is currently made by applying CP to voted labels from $\mathbb{P}_{\text{vote}}^{Y|X}$. In contrast, we make the usually implicit assumptions on label collection explicit. To perform calibration with $\mathbb{P}_{\text{agg}}^{Y|X}$, we propose a sampling-based approach, coined **Monte Carlo CP**, which proceeds by sampling multiple "pseudo ground truth" labels from $\mathbb{P}_{\text{agg}}^{Y|X}$ for each calibration example $X_1, \ldots, X_n$. We show how this approach can provide rigorous coverage guarantees despite not having access to exchangeable calibration examples. We present experiments on a skin condition classification case study: Here, calibration with voted labels from $\mathbb{P}_{\text{vote}}^{Y|X}$ is shown to undercover expert annotated labels by a staggering 10%. Our approach closes this gap both theoretically and empirically. Moreover, we discuss extensions to multi-label classification and calibration with augmented calibration examples for robust CP.

**Outline:** The rest of this paper is structured as follows: In Section 2, we illustrate the problem on a toy dataset and introduce required background on CP. Then, Section 3 introduces our original Monte Carlo CP procedures, highlights the applicability of this approach to related problems and discusses related work. In Section 4, we present an application to skin condition classification following (Liu et al., 2020), multi-label classification and data augmentation before concluding in Section 5.

## 2 Background and Motivating Example

### 2.1 Conformal prediction

In the following, we briefly review standard (split) CP (Vovk et al., 2005; Papadopoulos et al., 2002). To this end, we assume a classifier $\pi_y(x) \approx \mathbb{P}(Y = y|X = x)$ approximating the posterior label probabilities is available. This model will be typically based on learned parameters using a training set. Then, given a set of calibration examples $(X_i, Y_i)_{i\in[n]}$ from $\mathbb{P}$, we want to construct a prediction set $C(X) \subseteq [K]$ for the test point $(X, Y)$ such that the coverage guarantee from Equation (1) holds. As mentioned earlier, this requires the calibration examples and the test example to be exchangeable but does not make any further assumption on the data distribution or on the underlying model $\pi$.

A popular conformal predictor proceeds as follows: given a real-valued conformity score $E(X, k)$ based on the model predictions $\pi(x) \in \Delta^K$, we define

$$C(X) = \{k \in [K] : E(X, k) \geq \tau\} \tag{2}$$

where $\tau$ is the $\lfloor \alpha(n+1) \rfloor$ smallest element of $\{E(X_i, Y_i)\}_{i\in[n]}$, equivalently $\tau$ is obtained by computing the $\lfloor \alpha(n+1) \rfloor / n$ quantile of the distribution of the conformity scores of the calibration examples

$$\tau = Q\left(\{E(X_i, Y_i)\}_{i\in[n]}; \frac{\lfloor \alpha(n+1) \rfloor}{n}\right). \tag{3}$$

Here, $Q(\cdot; q)$ denotes the $q$-quantile. By construction of the quantile, see e.g. (Romano et al., 2019; Vovk et al., 2005; Angelopoulos & Bates, 2021), this ensures that the lower bound on coverage in Equation (1) is satisfied. Additionally, if the conformity scores are almost surely distinct, then we have the following upper bound $\mathbb{P}(Y \in C(X)) \leq 1 - \alpha + \frac{1}{n+1}$. The conformity score is a design choice. A standard choice, which we will use throughout the paper, is $E(x, k) = \pi_k(x)$ (Sadinle et al., 2019) but many alternative scores have been proposed in the literature (Romano et al., 2020; Angelopoulos et al., 2021).

[1] As explained further, we cannot obtain i.i.d. samples from $\mathbb{P}_{\text{agg}}^{Y|X}$ but only exchangeable ones. This prevents the use of concentration inequalities to obtain bounds on $\mathbb{P}_{\text{agg}}(Y \in C(X)|X = x)$ for a given $C(X)$.